

Ayush Maheshwari

Senior Software Developer

Skills

- Machine Learning
- Model Optimization
- Data Cleaning & Transformation
- PyTorch
- AWS
- Deep Learning
- CI/CD
- Batch Processing
- SQL
- Data visualization
- Orchestration
- Model Training & Evaluation
- Data Preprocessing
- Python
- GIT

Professional Experience

Senior Software Developer
Software Developer
Associate Software Developer

June 2025 - Present
March 2023 - May 2025
August 2021 - February 2023

Projects

Predictive Customer Churn Analytics Platform

- A telecom client required a predictive analytics platform to identify customers at risk of churn and enable proactive retention strategies. The objective was to develop scalable ML models with real-time scoring capabilities and deploy them using robust MLOps pipelines on cloud infrastructure.
- Developed and trained classification models (XGBoost, Random Forest) to predict customer churn with high accuracy.
- Performed advanced feature engineering using customer usage patterns, billing history, and interaction data.
- Built end-to-end MLOps pipelines using MLflow and Airflow for model training, versioning, and deployment.
- Deployed models as REST APIs using Django for real-time churn prediction.
- Integrated model monitoring to track drift, performance degradation, and retraining triggers.
- Collaborated with data engineers to build scalable data pipelines using AWS S3 and Redshift.
- Optimized model performance using hyperparameter tuning and cross-validation techniques.
- Designed dashboards to visualize churn trends and model predictions for business stakeholders.
- Enabled batch and real-time inference for marketing campaign integration.

Fraud Detection System for Digital Payments

- A fintech organization required a real-time fraud detection system to identify suspicious transactions and reduce financial risk. The goal was to build highly accurate ML models with low latency and deploy them in a scalable cloud environment.
- Built anomaly detection and classification models using Logistic Regression, Isolation Forest, and Neural Networks.
- Engineered transaction-based features such as velocity, geolocation, and behavioral patterns.

- Designed CI/CD pipelines for ML models using Docker for automated deployment.
- Deployed models on AWS SageMaker for scalable inference and monitoring.
- Performed model evaluation using precision-recall metrics to handle class imbalance.
- Implemented feature stores for reusable and consistent feature engineering across models.
- Collaborated with cross-functional teams to integrate fraud detection APIs into payment systems.
- Built alerting mechanisms for high-risk transactions with automated escalation workflows.

Personalized Recommendation Engine for E-commerce

- An e-commerce platform aimed to improve user engagement and conversion rates by implementing a personalized recommendation system. The solution focused on building scalable ML models and deploying them for real-time and batch recommendations.
- Developed collaborative filtering and deep learning-based recommendation models.
- Performed feature engineering on user behavior, purchase history, and product metadata.
- Built hybrid recommendation systems combining content-based and collaborative approaches.
- Implemented real-time recommendation APIs integrated with frontend applications.
- Optimized model performance using embeddings and neural network tuning techniques.
- Built analytics pipelines to track recommendation effectiveness and user interactions.

Demand Forecasting System for Supply Chain Optimization

- A retail company required a demand forecasting system to optimize inventory management and reduce stockouts. The project focused on building time-series forecasting models and deploying them with automated retraining pipelines.
- Developed time-series forecasting models using ARIMA, Prophet, and LSTM networks.
- Engineered features using historical sales data, seasonality, promotions, and external factors.
- Built automated MLOps pipelines for model retraining and performance monitoring.
- Deployed forecasting models on cloud platforms for scalable batch predictions.
- Implemented model evaluation metrics such as RMSE and MAPE for accuracy tracking.
- Designed workflows for continuous data ingestion and preprocessing using Airflow.
- Collaborated with business teams to align forecasts with supply chain planning.
- Integrated forecasting outputs with ERP systems for automated inventory decisions.
- Improved forecast accuracy through hyperparameter tuning and ensemble modeling.

Intelligent Document Classification and NLP Pipeline

- A legal-tech firm required an AI-driven document classification system to automate document processing and categorization. The objective was to leverage NLP models and deploy scalable pipelines for high-volume document processing.
- Built NLP-based classification models using BERT and TF-IDF with traditional ML algorithms.
- Performed text preprocessing, tokenization, and feature extraction for document data.
- Designed feature engineering pipelines for structured and unstructured text data.
- Implemented MLOps pipelines using MLflow for experiment tracking and deployment.
- Deployed models as microservices using Docker.
- Integrated OCR systems for document digitization and preprocessing.
- Evaluated models using accuracy, F1-score, and confusion matrix metrics.
- Collaborated with data scientists and backend engineers for seamless deployment.
- Built scalable pipelines to process large volumes of documents in near real-time.